Solar Si appendix

Sun · Si · downloadable appendix

products

No adopted Solar silicon abundance is available from this feed.

Highlighted Products

VIS

7.489 ± 0.074

±0.020 stat · ±0.071 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

7.577 ± 0.127

±0.036 stat · ±0.121 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

7.522 ± 0.069

±0.026 stat · ±0.064 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

7.325 ± 0.104

±0.035 stat · ±0.098 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · CRIRES+ H · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	38 products -- Si I · Asplund et al. 2021 lines; Si I · Deshmukh et al. 2022 lines; Si I · Elgueta et al. 2026 lines; Si II · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	24 products -- Si I · Codex-graded lines; Si II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS
A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.535 ±0.142 stat ±0.098 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.511 ±0.024 stat ±0.100 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.514 ±0.047 stat ±0.065 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.487 ±0.048 stat ±0.072 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.514 ±0.047 stat ±0.065 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.487 ±0.048 stat ±0.072 syst

VIS — IAG FTS
A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.519 ±0.094 stat ±0.103 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.489 ±0.105 stat ±0.101 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.511 ±0.015 stat ±0.070 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.484 ±0.016 stat ±0.076 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.511 ±0.015 stat ±0.070 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.484 ±0.016 stat ±0.076 syst

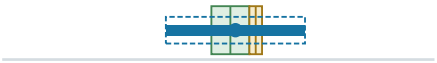
VIS — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.528 ±0.144 stat ±0.102 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.491 ±0.311 stat ±0.101 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.513 ±0.020 stat ±0.065 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.486 ±0.021 stat ±0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.513 ±0.020 stat ±0.065 syst

VIS — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.15–7.83

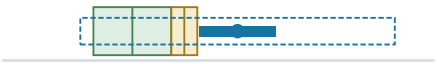
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.486 ±0.021 stat ±0.071 syst

VIS — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.15–7.83

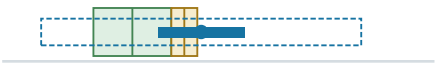
Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.518 ±0.110 stat ±0.110 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.488 ±0.285 stat ±0.104 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.516 ±0.019 stat ±0.064 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.489 ±0.020 stat ±0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.516 ±0.019 stat ±0.064 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.489 ±0.020 stat ±0.071 syst

VIS axis: 7.15 · 7.32 · 7.49 · 7.66 · 7.83 A(Si) dex

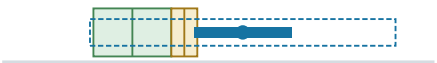
red-optical — IAG FTS
A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.591 ±0.030 stat ±0.121 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.573 ±0.028 stat ±0.125 syst

red-optical — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.41–7.74

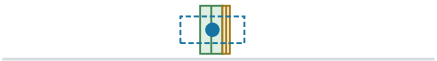
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.581 ±0.035 stat ±0.120 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.563 ±0.033 stat ±0.123 syst

red-optical — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.595 ±0.038 stat ±0.118 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.577 ±0.036 stat ±0.121 syst

red-optical axis: 7.41 · 7.49 · 7.58 · 7.66 · 7.74 A(Si) dex

NIR — CRIRES+ J
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 2 lines		7.513 ±0.005 stat ±0.086 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.487 ±0.009 stat ±0.091 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 27 lines		7.527 ±0.061 stat ±0.083 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 27 lines		7.507 ±0.061 stat ±0.088 syst

NIR — CRIRES+ Y
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.018 stat ±0.064 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.018 stat ±0.071 syst
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 2 lines		7.457 ±0.011 stat ±0.082 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.430 ±0.002 stat ±0.087 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		7.582 ±0.073 stat ±0.086 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 4 lines		7.560 ±0.078 stat ±0.091 syst

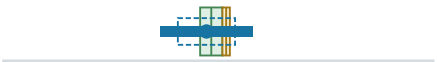
NIR — CRIRES+ Y (full arm)
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.018 stat ±0.064 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.018 stat ±0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 8 lines		7.551 ±0.057 stat ±0.062 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 8 lines		7.523 ±0.059 stat ±0.069 syst

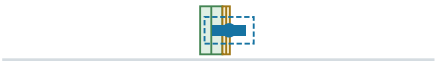
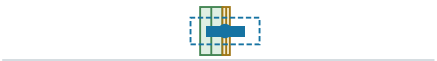
NIR — IAG FTS
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.554 ±0.121 stat ±0.070 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.542 ±0.121 stat ±0.076 syst

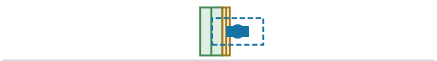
NIR — IAG FTS
A(Si) dex; axis 6.95–8.11

Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.528 ±0.088 stat ±0.135 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.497 ±0.125 stat ±0.077 syst

NIR — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.558 ±0.098 stat ±0.145 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.546 ±0.098 stat ±0.148 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.558 ±0.044 stat ±0.066 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.547 ±0.052 stat ±0.093 syst

NIR — Kitt Peak (1984, molecfit)
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 3 lines		7.581 ±0.030 stat ±0.069 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.552 ±0.035 stat ±0.075 syst
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 15 lines		7.536 ±0.027 stat ±0.057 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 15 lines		7.522 ±0.026 stat ±0.064 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 46 lines		7.536 ±0.033 stat ±0.515 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 43 lines		7.529 ±0.035 stat ±0.552 syst

NIR axis: 6.95 · 7.24 · 7.53 · 7.82 · 8.11 A(Si) dex

H — CRIRES+ H
A(Si) dex; axis 7.20–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.363 ±0.038 stat ±0.093 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.325 ±0.035 stat ±0.098 syst

H — CRIRES+ H (full arm, telluric-corrected)
A(Si) dex; axis 7.20–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.418 ±0.054 stat ±0.108 syst

H — CRIRES+ H (full arm, telluric-corrected)
A(Si) dex; axis 7.20–7.59

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 3 lines



7.380
±0.047 stat ±0.112 syst

H axis: 7.20 · 7.30 · 7.39 · 7.49 · 7.59 A(Si) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every silicon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the silicon abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
VIS	Si I · Codex-graded lines; Si II · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
VIS	Si I · Asplund et al. 2021 lines; Si II · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
red-optical	Si I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	7.563–7.577 (1D-NLTE)	line strengths (gf) 0.11, fit / line-to-line scatter 0.04
NIR	Si I · Codex-graded lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	4–46	7.497–7.560 (1D-NLTE)	telluric residual 0.55, line strengths (gf) 0.04
NIR	Si I · Deshmukh et al. 2022 lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	7.428–7.552 (1D-NLTE)	line strengths (gf) 0.06, fit / line-to-line scatter 0.04
NIR	Si I · Elgueta et al. 2026 lines	CRIRES+ Y, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–15	7.430–7.546 (1D-NLTE)	line strengths (gf) 0.04, fit window 0.03

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
H	Si I · Elgueta et al. 2026 lines	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)	1D-LTE, 1D-NLTE	3	7.325–7.380 (1D-NLTE)	continuum placement 0.06, line strengths (gf) 0.05

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 59 lines.

Abundance outlier: disagrees with the other lines well beyond its error, and no solar analysis in the literature uses it.

Not measurable: the line falls in a data gap or a saturated telluric band.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
Si II	3203.872	saturated (log W/λ = -4.18)
Si II	3210.028	saturated (log W/λ = -3.98)
Si II	3339.818	saturated (log W/λ = -4.04)
Si II	3856.017	saturated (log W/λ = -3.90)
Si II	3862.595	saturated (log W/λ = -4.15)
Si I	3905.523	saturated (log W/λ = -3.85)
Si I	4102.936	saturated (log W/λ = -4.10)
Si I	5948.541	saturated (log W/λ = -4.79)
Si I	7918.383	saturated (log W/λ = -4.88)
Si I	7932.348	saturated (log W/λ = -4.81)
Si I	7944.001	saturated (log W/λ = -4.75)
Si I	8543.151	saturated (log W/λ = -4.23)
Si I	8648.465	saturated (log W/λ = -4.72)
Si I	8661.177	saturated (log W/λ = -4.27)
Si I	8728.010	saturated (log W/λ = -4.89)
Si I	8742.446	saturated (log W/λ = -4.90)
Si I	8752.007	saturated (log W/λ = -4.79)
Si I	8790.389	saturated (log W/λ = -4.82)
Si I	9021.590	saturated (log W/λ = -4.88)
Si I	9413.500	saturated (log W/λ = -4.56)
Si I	9887.047	abundance outlier
Si I	10313.197	abundance outlier
Si I	10371.263	saturated (log W/λ = -4.73)
Si I	10585.141	saturated (log W/λ = -4.46)
Si I	10603.425	saturated (log W/λ = -4.57)
Si I	10627.648	saturated (log W/λ = -4.85)
Si I	10660.973	saturated (log W/λ = -4.53)
Si I	10689.716	saturated (log W/λ = -4.70)
Si I	10694.251	saturated (log W/λ = -4.66)
Si I	10727.406	saturated (log W/λ = -4.62)

Species	Wavelength (Å, air)	Why it is excluded
Si I	10749.378	saturated (log W/λ = -4.54)
Si I	10786.849	saturated (log W/λ = -4.59)
Si I	10827.088	saturated (log W/λ = -4.40)
Si I	10843.858	saturated (log W/λ = -4.73)
Si I	10868.790	saturated (log W/λ = -4.49)
Si I	10869.536	saturated (log W/λ = -4.34)
Si I	10885.333	saturated (log W/λ = -4.84)
Si I	10979.308	saturated (log W/λ = -4.62)
Si I	10982.058	saturated (log W/λ = -4.90)
Si I	11017.966	saturated (log W/λ = -4.68)
Si I	11187.588	saturated (log W/λ = -4.89)
Si I	11289.829	saturated (log W/λ = -4.86)
Si I	11611.060	saturated (log W/λ = -4.60)
Si I	11984.198	saturated (log W/λ = -4.46)
Si I	11991.568	saturated (log W/λ = -4.51)
Si I	12031.504	saturated (log W/λ = -4.38)
Si I	12103.534	saturated (log W/λ = -4.59)
Si I	12189.241	abundance outlier
Si I	12270.692	saturated (log W/λ = -4.62)
Si I	15077.332	saturated (log W/λ = -4.67)
Si I	15557.778	saturated (log W/λ = -4.88)
Si I	15888.409	saturated (log W/λ = -4.39)
Si I	15960.063	saturated (log W/λ = -4.53)
Si I	16060.009	saturated (log W/λ = -4.64)
Si I	16163.691	saturated (log W/λ = -4.78)
Si I	16380.176	saturated (log W/λ = -4.72)
Si I	16381.534	saturated (log W/λ = -4.54)
Si I	16434.927	not measurable
Si I	17466.879	saturated (log W/λ = -4.45)

Solar silicon context

Silicon is traced by Si I lines from the blue to the H band and by the single Si II 6371 Å line. The optical lines are Asplund et al. 2021's set (Amarsi & Asplund 2017) with the gf the authors used; the infrared lines are the published sets of Deshmukh et al. 2022 and Elgueta et al. 2026 on Pehlivan Rhodin et al. 2024 oscillator strengths. Each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Si appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit ccc31e67aa934991efdb45b6be7c246c8a7d733c
Feed Si.json v1.278 · updated 2026-10-17T23:20:32Z
Generated 2026-10-17T17:26:35-06:00 (source commit timestamp)
Feed SHA-256 7c1195f65f701ad6d0d6dbd97f55dee5cfdd7b07b6f7a1291b6ab2ce2619f708
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Si comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set SI_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Si II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [6].

Solar abundance and indicator analyses. Published line set SI_ELGUETA2026 (the line selection). Sources: [8].

Atomic data and gf authorities. Si I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Si_Amarsi2020_PySME.csv). Sources: [10].

Solar abundance and indicator analyses. Published line set SI_DESHMUKH2022 (the line selection). Sources: [11].

Product-to-source index

VIS · HARPS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-18313ebf6e74

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